What is **Azure Active Directory (Azure AD)**, and how does it differ from on-premises **Active Directory**?

Ans : It is an identity and access management service help employees to sign in and access resources securely.

Azure AD : Run on cloud, easily scalable, manage internal as well as external users.
On premises AD : run on premises, limited scalability, manage only internal users.

Explain **Azure AD B2B vs. Azure AD B2C** with real-world examples.

Ans :
B2B : Collaborate with partners, vendors, and suppliers by allowing them access to internal resources. Uses the guest user's identity. Limited customization of login experience.

B2C : Manage customer identities and enable secure sign-up, sign-in, and profile management. Users sign up with social accounts. Fully customizable login UI/flows

Create a list of at least **five authentication methods** available in Azure AD.

Ans :
1. **Standard Password** : User have to enter username and password to verify
2. **MFA** : User need to provide two or more forms to verify. It can be Biometrics, Microsoft Authenticator app.
3. **SSO** : User can once sign in and can access the services w/o sign in again
4. **Password less** : The user need to provide the mail ID, in response to that the user get an OTP, through which he need to verify.
5. **Certificate based authentication** : The user must have a certificate to verify themselves.

What is **Multi-Factor Authentication (MFA)**, and why is it important?

Ans: MFA is a security process where users must provide two or more forms of verification to access an account or app.

Example : Microsoft Authenticator App, Biometrics.

Why is it important?

- Reduces the risk of unauthorized access.
- Even if a password is stolen, the attacker still needs a second factor.
- Essential for securing sensitive data and apps.

Define **Role-Based Access Control (RBAC)** and give an example of how it's used ?

Ans: RBAC is a method of restricting access based on a user's role in an organization.

Key components:

- **Roles:** Define what actions can be performed (e.g., read-only, contributor, admin).
- **Permissions:** Specific actions allowed by a role.
- **Assignments:** Users or groups assigned to roles.